

Improved Productivity Using Deep Learning-Assisted Reporting for Lumbar Spine MRI

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Inventory Category:	DL stenosis / multi-pathology
Comparability / Priority:	High Priority: Critical

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Reader-assistance / workflow study Lumbar MRI
Dataset & Cohort:	nan (Sample: nan)
Disease / Target:	Lumbar spinal stenosis reporting
Model / AI Method:	DL decision support
Evaluation Metrics:	Reporting time, interobserver agreement, diagnostic performance
Key Finding / Relevance:	DL assistance reduced reporting time and maintained or improved agreement.
Thesis Context (Ch 2):	Connects technical performance with actual clinical workflow value.

Abstract / Summary

Background Lumbar spine MRI studies are widely used for back pain assessment. Interpretation involves grading lumbar spinal stenosis, which is repetitive and time consuming. Deep learning (DL) could provide faster and more consistent interpretation. Purpose To assess the speed and interobserver agreement of radiologists for reporting lumbar spinal stenosis with and without DL assistance. Materials and Methods In this retrospective study, a DL model designed to assist radiologists in the interpretation of spinal canal, lateral recess, and neural foraminal stenoses on lumbar spine MRI scans was used. Randomly selected lumbar spine MRI studies obtained in patients with back pain who were 18 years and older over a 3-year period, from September 2015 to September 2018, were included in an internal test data set. Studies with instrumentation and scoliosis were excluded. Eight radiologists, each with 2–13 years of experience in spine MRI interpretation, reviewed studies with and without DL model assistance with a 1-month washout period. Time to diagnosis (in seconds) and interobserver agreement (using Gwet κ) were assessed for stenosis grading for each radiologist with and without the DL model and compared with test data set labels provided by an external musculoskeletal radiologist (with 32 years of experience) as the reference standard. Results Overall, 444 images in 25 patients (mean age, 51 years $\pm$ 20 [SD]; 14 women) were evaluated in a test data set. DL-assisted radiologists had a reduced interpretation time per spine MRI study, from a mean of 124–274 seconds (SD, 25–88 seconds) to 47–71 seconds (SD, 24–29 seconds) ($P < .001$). DL-assisted radiologists had either superior or equivalent interobserver agreement for all stenosis gradings compared with unassisted radiologists. DL-assisted general and in-training radiologists improved their interobserver agreement for four-class neural foraminal stenosis, with κ values of 0.71 and 0.70 (with DL) versus 0.39 and 0.39 (without DL), respectively (both $P < .001$). Conclusion Radiologists who were assisted by deep learning for interpretation of lumbar spinal stenosis on MRI scans showed a marked reduction in reporting time and superior or equivalent interobserver agreement for all stenosis gradings compared with radiologists who were unassisted by deep learning. © RSNA, 2022 Online supplemental material is available for this article. See also the editorial by Hayashi in this issue.